

Adriel Bobby

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Certificate Drive: Available upon request

Education	Institute/School	Year	CGPA
B.Tech., Computer Science Engineering	Rajagiri School of Engineering and Technology, Kochi	2024-28	8
Class 12, CBSE	Rajagiri Public School, Kalamssery	2022-24	8.9
Class 10, CBSE	The Charter School, Pukattupady	2021-22	8.9

CAREER OBJECTIVE

Computer Science Engineering student with a passion for cybersecurity, digital communication, and social media management. Aspiring cybersecurity specialist with strong interests in ethical hacking and cyber-threat analysis.

PROFESSIONAL EXPERIENCE

- **Cybersecurity Intern** | *Kerala Police Cyberdome, Trivandrum* (2 May 2026 – 2 Jun 2026)
 - Performed web application security testing, identifying an authentication bypass and XSS in session parameters on internal applications.
 - Set up an Android malware analysis lab with Genymotion, MobSF, Frida, and ADB; investigated a suspected malicious challan app that used anti-VM checks and PII-harvesting forms to evade automated analysis.
 - Strengthened networking fundamentals by studying OSI/TCP/IP models, VLANs, NAT, subnetting, IPv4/IPv6, and packet flow from local host to internet services and documented common network troubleshooting commands and packet lifecycles to support future labs and security investigations.
 - Explored containerization concepts with Docker and learned common web attack vectors such as SQL injection and CSRF.

CERTIFICATES

- **Certified Penetration Tester** | *RedTeam Academy* (May'25 - August'25)
 - Gained hands-on experience in ethical hacking, vulnerability scanning, and system exploitation.
 - Used industry tools like Nmap, Burp Suite, and Metasploit.
 - Practiced real-world attack scenarios including privilege escalation and post-exploitation.
 - Learned structured documentation and reporting of penetration test findings.

TECHNICAL SKILLS

- **Programming & Scripting Languages:** C, Python, C++, Java, JavaScript
- **Security & Analysis Tools:** Linux, Nmap, Gobuster, WPScan, Joomscan, SQLMap, Metasploit, Searchsploit, Hydra, John the Ripper, Hashcat, Netcat, Burp Suite, Wireshark, Tor, Steghide, Stegseek, MobSF, Frida, Genymotion, ADB, VirtualBox, Docker
- **Databases & Platforms:** MySQL, pfSense, Kali Linux, Ubuntu Server
- **Tech Domains:** Web Application Security, Network Security, Mobile Security, Social Media Management

POSITIONS OF RESPONSIBILITY

- **Electronic Communications Coordinator** | *IEEE RSET Student Branch* (March'25-March'26)
 - Responsible for handling all official announcements and communication for the Student Branch.
- **Technical Coordinator** | *IEEE RSET Student Branch* (March'26-Present)
 - Supporting the technical side of IEEE RSET SB events - setting up hardware demos, assisting speakers with lab equipment, and helping run hands-on workshops.

MAJOR PROJECTS

- **Vaccine Dispatch Management System**
 - Built a Python + MySQL application to manage vaccine inventory, hospital orders, and dispatch operations for a pharmaceutical workflow. [page:2]
 - Implemented admin-protected vaccine entry, order placement by hospital and state, dispatch logging with dates, and pending-order tracking. [page:2]
 - Added sales and revenue reporting with matplotlib bar-chart visualizations to highlight dispatch volume and profitability per vaccine. [page:2]
 - Refactored the codebase into separate database and business-logic modules for easier maintenance and reuse.
 - **Tech Used:** Python, MySQL, CustomTkinter, Matplotlib, Virtualenv

- **Mis-Communication-Nater**
 - Developed an experimental AI system that demonstrates the concept of artificial stupidity through intentional design flaws.
 - Uses two AI models that communicate with each other rather than answering queries, creating a deliberately non-functional output.
 - Highlights inefficient use of compute as commentary on AI development and resource allocation.
 - **Tech Used:** Raspberry Pi 4 Model B, Raspberry OS Lite, StableLM Zephyr 3B.Q2_K, LEDs, Web UI
- **ESP32 Marauder (Wi-Fi and Bluetooth Pentesting Tool)**
 - Setting up and customizing the ESP32 Marauder firmware for wireless auditing and pentesting.
 - Exploring Wi-Fi deauthentication, packet sniffing, and Bluetooth scanning capabilities.
 - Aims to deepen hands-on understanding of hardware-based network attacks.
 - **Tech Used:** ESP32, Arduino IDE, Marauder Firmware, Wireshark
- **Homelab Environment for Cybersecurity Practice**
 - Building a self-hosted homelab to simulate enterprise network environments.
 - Deploying virtual machines and vulnerable boxes for penetration testing and tool practice.
 - Focused on improving workflow with tools like Nmap, Metasploit, Burp Suite, and SIEMs.
 - **Tech Used:** VirtualBox, Ubuntu Server, Kali Linux, pfSense, Docker
- **PoolDetect AI – High-Speed Satellite Pool Detection**
 - Designed an end-to-end pipeline that converts zip codes into satellite tiles and performs automated pool detection and property intelligence.
 - Replaced heavy YOLO-style models with a custom OpenCV engine using morphological operations and heuristics for sub-0.1s per-image processing on CPU-only infrastructure.
 - Implemented multi-layer false-positive filtering (water body size checks, tennis-court geometry, blue-roof spectral analysis, pool-shape circularity and solidity) to significantly boost precision.
 - Added four-way pool classification and change-detection analytics to identify new pool constructions and estimate property value from SVG polygon areas.
 - **Tech Used:** Python, OpenCV, NumPy, Nominatim API, Esri World Imagery
- **Calm-Cockpit – Minimal Distraction Monitoring Dashboard**
 - Designed a dashboard that surfaces only the most important system and status signals, reducing noise and helping operators focus on critical changes.
 - Implemented a clean visual layout with prioritized alerts and color cues, inspired by calm-technology and cockpit-style UI principles.
 - **Tech Used:** Python, Flask, SQL, JS, CSS Grid
- **Gemini Context Monitor**
 - Built a tool to visualize and track how long-context AI models use conversation history, highlighting when older messages fall out of effective context.
 - Logged prompts and responses to analyze token usage and model behavior over time, helping debug prompt design and context-window limits.
 - **Tech Used:** Python, CustomTkinter, SQLite, Rich, Pydantic
- **Student Marks Tracker – Internship Performance Dashboard**
 - Developed a web application for tracking internship performance, allowing incharges to register students, define internship date ranges, and record daily marks with qualitative remark bands (Terrible–Excellent). [page:3]
 - Implemented logic for internship windows, holiday-aware mark entry (skipping Sundays and 2nd Saturdays), and automatic visibility of only active students on a given day. [page:3]
 - Built aggregate scorecards per student, showing average marks, working-day coverage, and distribution of performance bands, backed by an SQLite database accessed through Prisma. [page:3]
 - **Tech Used:** Next.js, TypeScript, SQLite, Prisma ORM, Vanilla CSS

HACKATHON SUBMISSIONS

- **High Torque Vertical Axis Wind Turbine Street Lamp** *3rd Price*
 - The project involves the development of a **High Torque Vertical Axis Wind Turbine (VAWT) Street Lamp** system.
 - The system uses a VAWT to convert wind energy into mechanical rotation, which then powers a generator to produce electricity. This electricity is stored in a battery and used to power an LED streetlight.
 - The key components include the VAWT, generator, battery storage, LDR control system, and LED streetlight. The design aims to automate the streetlight operation based on ambient light levels.
 - The initial low starting torque issue was addressed with a kickstarter system, and the over-speeding issue was resolved using a pitch control system.
- **KruizeX Ideathon – Smart Multimodal Transit Queue System** *1st Price*
 - Proposed a digital queue and real-time tracking solution to reduce boarding uncertainty and missed connections in Kochi Water Metro and feeder buses.
 - Designed a token-based passenger numbering system with display screens to clearly indicate boarding order and improve commuter experience.
 - Recommended integrating live boat and bus tracking into the Kochi One platform to provide ETAs, delay notifications, and seamless rail–water–bus journey planning.

REFERENCES

- **Academic Reference:**
Dr. Jomina John
Assistant Professor, Department of Computer Science and Engineering
Rajagiri School of Engineering and Technology, Kochi
Contact details available upon request.

SELF DECLARATION

I hereby declare that all the information provided above is true and correct to the best of my knowledge and belief.